

Manqin Zeng

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📍 Shenzhen, Guangdong, China



EDUCATION

The Chinese University of Hong Kong

Research Assistant

- **Advisor:** Prof. Li Zhang (IEEE Fellow)
- **Research Area:** Wrist-fingertip collaborative tactile perception; Iontronic tactile sensing

May 2026 – Present
Hong Kong, China

Southern University of Science and Technology & Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

Master's Degree (Joint Program) | Expected graduation: June 2027

- **Advisors:** Prof. Xinyu Wu (IEEE Fellow), Prof. Zhengkun Yi
- **Research Area:** Tactile sensing technology and multimodal fusion

Sept. 2024 – Present
Shenzhen, China

South China University of Technology

Bachelor's Degree | Robotics Engineering (English-Taught) | GPA: 3.63/4.0

- **Advisor:** Prof. Yong Zhong
- **Outstanding Bachelors Thesis (Top 5%):** Automated Material Exchange System for Soft Robotic 3D Printing
- **Core Coursework:** AI Tech & Applications (96), Data Structures & Algorithms (96), Mechatronics (99), Classical Control Theory (90), Mechanics of Materials (96), Dynamic System Modeling (99)
- **National Scholarship (2022-2023), University Outstanding Student Club Leader**

Sept. 2020 – July 2024
Guangzhou, China

RESEARCH EXPERIENCE

Bio-inspired Robotic Sensors and Multimodal Fusion Networks | *2 conference papers*

2024 – Present

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences & The Chinese University of Hong Kong

Advisors: Prof. Xinyu Wu, Prof. Li Zhang

- **Event-Frame Dual-Modal Tactile Sensor:** Engineered a bio-inspired optical tactile sensor utilizing a DAVIS346 camera to replicate the human fast-slow tactile pathways, achieving microsecond-resolution dynamic event streams and grayscale frames with precise spatio-temporal alignment.
- **Multimodal Fusion Network:** Investigated data fusion strategies for event and frame modalities across early, mid, and late stages, outperforming the best single-modal baseline by 5%–8% on two challenging 10-class tactile recognition tasks: stiffness classification of 10 polyurethane foams and texture recognition of 10 diverse fabrics.
- **State Recognition Networks for Robotic Assembly:** Developed a residual Bi-LSTM network with event-based sparse Transformer to accurately classify diverse snap-fit engagement states, automatically identifying assembly successes or failures.
- **Biomimetic Iontronic Tactile System:** Integrated flexible iontronic sensing materials onto a ROBOTIQ gripper to mimic human skin's tactile perception, enabling dexterous physical interaction and adaptive grasping during robotic assembly tasks.

Automated Material Exchange System for Soft Robotic 3D Printing | *Outstanding Bachelor's Thesis*

2023 – 2024

South China University of Technology

Advisor: Prof. Yong Zhong

- **Automated Material Exchange:** Engineered an integrated three-nozzle syringe-extrusion mechanism for automated material switching and continuous deposition of gel-based components in a support matrix.
- **Extrusion-Based Print Head:** Designed a lightweight extrusion print head with optimized pillar-supported dual-ring fixation, significantly reducing payload and operational complexity for multi-nozzle integration.

Smart Kitchen Equipment utilizing Pulsed Electric Field | *1 journal paper, 1 patent*

2020 – 2021

South China University of Technology & Midea

Advisor: Prof. Zhong Han

- **PEF-Assisted Meat Tenderization:** Developed an automated pulsed electric field system integrated with ultrasound to disrupt muscle fiber Z-lines, significantly tenderizing beef and reducing harmful byproduct release for enhanced safety.
- **PEF-Induced Protein Modification:** Applied synergistic pulsed electric field and pH-shifting treatments to unfold soy protein isolate structures, successfully preparing nanoparticles with a 41% higher lutein loading capacity.
- **Human-Machine Collaboration:** Designed an intelligent pre-processing station for kitchen collaboration, autonomously handling time-consuming preparation to save labor and standardize workflows.

PUBLICATIONS

- [1] **Manqin Zeng**, Bozhan Cao, Senlin Fang, et al., "Bio-inspired Event-frame Dual-modal Tactile Sensor for Texture and Stiffness Classification," *IROS (IEEE International Conference on Robotics and Biomimetics) 2026, Pittsburgh, America*.
- [2] **Manqin Zeng**, Chengliang Liu, Bozhan Cao, et al., "SF-Net: A Force-Based, Formula-Enhanced Residual Bi-LSTM for Snap-Fit State Classification," *ROBIO (IEEE/RSJ International Conference on Intelligent Robots and Systems) 2025, Chengdu, China*.
- [3] Rui Wang*, **Manqin Zeng***, Yu-Wei Wu, et al., "Enhanced encapsulation of lutein using soy protein isolate nanoparticles prepared by pulsed electric field and pH shifting treatment," *Food Chemistry 2023* (Co-first author).
- [4] Bozhan Cao, Senlin Fang, Zeqin Lin, **Manqin Zeng**, et al., "Event-Based Optical Tactile Sensor and Sparse Transformer for Robust Snap-Fit Recognition," *ROBIO (IEEE/RSJ International Conference on Intelligent Robots and Systems) 2025, Chengdu, China*.
- [5] **Manqin Zeng**, Zhong Han, Jinlin Cai et al., "Method and device for tenderizing and efficient marinating of beef by ultrasonic synergistic pulsed electric field." *Invention Patent CN 113796502 B*.

RESEARCH INTERESTS

Robotic Mechanism & Hardware Design: Focusing on the mechanical design, CAD modeling, and physical prototyping of novel sensors and robotic structures, with an emphasis on iterative hardware tuning and empirical validation.

Multimodal Learning & Data Processing: Developing neural network architectures to process and analyze multimodal sensory data collected from real-world experiments, enabling accurate classification, state recognition, and prediction.

3D Printing for Soft & Biomaterials: Innovating additive manufacturing strategies for unconventional materials (e.g., flexible and biomaterials), including the custom design of extrusion nozzles and specialized material-shaping systems.

COMPETITIONS

L'Oréal 2022 Brandstorm | *Mainland China Runner-up (2/13000), Global Top 40* | *Team Leader* 2022

- **3D Printed Facial Masks using Biomaterials:** Led the core technology development of a flexible 3D printer utilizing photo-cured seaweed hydrogel to produce fully customized facial masks.
- Received strong endorsement from Kiehl's technical advisors, leading to an institutional exchange at L'Oréal China Headquarters and an industrial field study at the L'Oréal Suzhou Factory.

DJI 21st RoboMaster National Robotics Competition | *National First Prize* | *South China Tiger Team Member* 2022

- **Structural Design of Projectile-Shooting Robotic Vehicles:** Contributed to the mechanical design and construction of custom ground vehicles designed to shoot projectiles and capture strategic resources in a highly dynamic combat arena.
- Competed as a core member of SCUT's South China Tiger team (a traditional powerhouse with multiple national titles) in this prestigious DJI-sponsored national robotics combat competition.

MoE "Internet+" National Innovation Competition (Startup Group) | *Provincial Silver, Univ. Gold* | *Co-Leader* 2022

- **PEF-Based Meat Pre-processing Equipment:** Developed a large-scale automated meat processing system leveraging patented pulsed electric field (PEF) technology.
- Successfully manufactured the physical equipment prototype with massive market potential, gaining high recognition from an academician's team and multiple food enterprises.

MoE "Internet+" National Innovation Competition (Idea Group) | *Univ. Silver* | *Co-Leader* 2022

- **Linux-Based First-Person RoboMaster Simulator:** Co-developed a comprehensive Linux-based first-person simulator for the RoboMaster competition to provide a realistic robotic operation and combat experience.
- Partnered officially with DJI to host a national simulator league (XBOTCON 2022) across 30+ universities, actively managing product packaging, rule design, and the roadshow defense.

INDUSTRY EXPERIENCE

KUKA China (Midea) Jul. 2021 – Aug. 2021
Research and Development Department | *Summer Intern* Foshan, China

- Assisted in the inbound testing and debugging of KUKA 6-axis robotic arms; gained proficiency in UR robotic arm operation and frontline industrial packaging processes.
- Facilitated cross-departmental communication by translating and summarizing technical meetings in English.

SKILLS & INTERESTS

Technical Skills: ROS, PyTorch, SolidWorks Modeling, 3D Printing, vibe coding

Languages & Writing: English (CET-4: 619, CET-6: 600), Professional Working Proficiency (English-taught Programs for B.Sc. & M.Sc.); ROBIO Oral Presenter

Interests: Social Media Operations (5000+ followers across platforms), Digital Art, Literature